moving in opposite direction gives rise to stationary wave. {B1]

6 (a)

$$R = \frac{\rho l}{A} = \frac{1.7\times10^{-8}\times1.5}{3.2\times10^{-9}} \quad [M1]$$

$= 8.0\ \Omega$ [A0]

6 (b)

$E = IR$

$12 = I \times (1.0 + 2.0 + 8.0)$

$I = 1.091\ A$ [C1]

$I = nAqv$

$1.091 = (8.5 \times 10^{28})(3.2 \times 10^{-9})(1.6 \times 10^{-19})\ v$

$v = 0.025\ m\ s^{-1}$ [A1]

6 (c)(i)

Using potential divider equation,

$$V_{PQ} = \frac{R_{RQ}}{R_{total}} = \frac{8.0}{8.0+1.0+2.0}\times12.0 \quad [M1]$$

$= 8.7\ V$ [A0]

6 (c)(ii)

$$\text{terminal pd of B} = \frac{R}{R+r}\times E = \frac{4.0}{4.0+3.0}\times1.5$$

$= 0.857\ V$ [C1]

$$\frac{V_{PJ}}{V_{PQ}} = \frac{l_{PJ}}{l_{PQ}}$$

Since terminal pd of B = V_{PJ}

$$\frac{0.857}{8.7} = \frac{l_{PJ}}{1.5} \quad [C1]$$

$l_{PJ} = 0.148\ m$ [A1]

6 (c)(iii)

To improve accuracy, the modification is either to

1. make the potential difference of PJ larger by increasing the resistance of the 4.0 Ω larger [B1].

However this amendment has minimal effect due to the small value of e.m.f. of cell B

Only award 1 mark.

2. make the potential difference per unit length of PQ smaller [M1] to increase the length of PQ needed at balance length by decreasing the e.m.f. of cell A or reducing the resistance of the wire through increasing its cross-sectional area, decreasing length or resistivity or increasing the resistance of the 2.0 Ω resistor [A1]

6 (d)

From galvanometer towards J

7 (a)